

Diagnostic Imaging of Lower Urinary Tract Disease



Silke Hecht, Dr med vet

KEYWORDS

- Contrast cystography and urethrography • Ultrasonography • CT/MRI
- Urinary bladder • Urethra • Transitional cell carcinoma • Cystitis • Urolithiasis

KEY POINTS

- Diagnostic imaging of the lower urinary tract is commonly performed in small animal patients presented with urinary tract signs such as azotemia, hematuria, dysuria, stranguria, and pollakiuria.
- Ultrasonography is the modality of choice for the diagnosis of most disorders affecting the bladder and intra-abdominal portion of the urethra.
- Survey radiographs of the abdomen are useful in identifying radiopaque bladder or urethral calculi, gas within the urinary tract, and enlarged lymph nodes or bone metastases secondary to lower urinary tract neoplasia.
- Cystography and urethrography are especially useful in the evaluation of bladder or urethral rupture, abnormal communication with other organs, and lesions of the pelvic or penile urethra.
- Advanced diagnostic imaging is useful in assessing the ureterovesical junction, evaluating intrapelvic lesions, improving surgical planning, monitoring the size of known lesions associated with the urinary tract, and evaluating regional lymph nodes and osseous structures for metastases.

INTRODUCTION

Diagnostic imaging of the lower urinary tract is routinely performed in small animal practice. This article reviews the indications, imaging modalities used, and findings to be expected in common lower urinary tract diseases.

INDICATIONS

Indications for diagnostic imaging of the lower urinary tract overlap with indications for general abdominal imaging (eg, nonspecific clinical signs such as anorexia or

The author has nothing to disclose.

Department of Small Animal Clinical Sciences, College of Veterinary Medicine, University of Tennessee, 2407 River Drive, Knoxville, TN 37996, USA

E-mail address: shecht@utk.edu

Vet Clin Small Anim 45 (2015) 639–663

<http://dx.doi.org/10.1016/j.cvsm.2015.02.002>

vetsmall.theclinics.com

0195-5616/15/\$ – see front matter © 2015 Elsevier Inc. All rights reserved.

abdominal pain and abnormal findings on abdominal palpation). Specific indications include azotemia, hematuria, dysuria, stranguria, pollakiuria, and abdominal trauma with suspicion of uroabdomen.

IMAGING MODALITIES AND TECHNIQUES USED FOR LOWER URINARY TRACT IMAGING

Survey Radiographs

As radiographic equipment is readily available in small animal practices as well as referral institutions, survey radiographs of the abdomen are commonly the first step used in the evaluation of patients presented with signs referable to the lower urinary tract.¹⁻⁴ They are useful in identifying radiopaque bladder or urethral calculi or gas within the urinary tract and in assessing a patient for enlarged lymph nodes or bone metastases secondary to lower urinary tract neoplasia.⁵⁻⁸ Other disorders of the lower urinary tract usually require additional imaging for diagnosis, although dystrophic mineralization, urethral thickening, or prostatomegaly may on occasion be visible in dogs with urinary tract neoplasia (see later discussion). Standard radiographic projections include lateral and ventrodorsal views. Care should be taken to always include the entire pelvis and perineal area to allow assessment of the pelvic urethra. In addition, ventrodorsal oblique views (**Fig. 1**) and in male dogs lateral views of the pelvis with the legs flexed forward (“lateral perineal views” or “butt shots”) may be needed to minimize superimposition of osseous structures, which may obscure urethral calculi (**Fig. 2**).

Normal findings

The normal urinary bladder is of homogenous fluid opacity, is variable in size and shape, and may be partially located within the pelvic canal. Intrapelvic location of the bladder may be noted, the significance of which is controversially discussed.^{9,10} The normal urethra is not visible on survey radiographs.⁴

Contrast Cystography and Urethrography

These procedures involve instillation of a positive and/or negative contrast agent into the urinary bladder and/or urethra.^{2,3,11-18} They have largely been replaced by abdominal ultrasound but are still indicated for certain conditions (see later discussion).

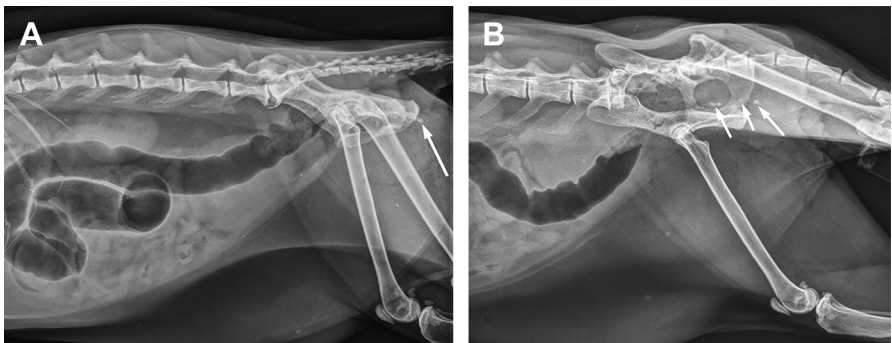


Fig. 1. Urethral calculi in a 13-year-old male castrated domestic shorthair cat. On the lateral view (**A**), an oblong mineral opacity is partially superimposed over the pelvis and extends into the plane of the perineal soft tissues (*arrow*). The oblique view of the pelvis (**B**) allows for identification of additional calculi in the intrapelvic part of the urethra (*arrows*). Mineralized sediment is present with the urinary bladder. Lumbosacral spondylosis deformans is incidental.

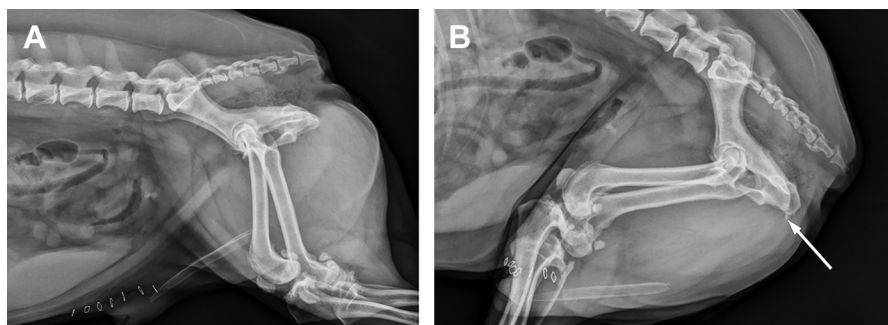


Fig. 2. Bladder and urethral calculi in an 8-year-old male castrated Pembroke Welsh corgi. The routine lateral view of the caudal abdomen (A) shows 2 sharply margined calculi in the plane of the urinary bladder. The lateral perineal view with the legs flexed forward (B) shows an additional urethral calculus immediately distal to the ischium (arrow). Skin staples associated with the caudal ventral abdominal wall are consistent with recent surgery, and bilateral stifle degenerative joint disease is incidental.

Although these techniques are generally considered safe, adverse effects may occur, including hematuria, infection, (hemorrhagic) cystitis or urethritis, dissection of contrast medium into the bladder wall, iatrogenic bladder or urethral rupture, knotting or breakage of the catheter, and air embolization if using room air as contrast agent.^{19–24} Urinalysis and urine culture should be performed before urinary contrast studies because of the risk of contamination during these procedures and because some results may be affected by prior contrast medium administration.^{25–27}

Negative contrast cystography

Negative contrast cystography using a gaseous contrast agent is uncommonly performed. CO₂ and N₂O have a higher solubility in blood than room air and therefore pose a lower risk of embolization if absorbed into the vascular system.^{2,14,23,24,28} Negative contrast cystography is useful in localizing the bladder if caudal abdominal masses are present or urinary bladder herniation is suspected. The technique is also useful in identification of luminal filling defects (bladder calculi) but is inferior to double-contrast cystography in this regard. It may be used as a complementary technique when performing an intravenous pyelogram because it allows delineation of caudal ureters and ureteral papillae.²⁹

Positive contrast cystography

This technique involves instillation of positive contrast medium into the urinary bladder and is the preferred method to diagnose a bladder rupture or abnormal communication with adjacent structures (eg, urethrorectal fistula).^{1,2,13,17,28,30–33} It is also useful in distinguishing the urinary bladder from caudal abdominal masses and in evaluating the position of the urinary bladder in the case of abdominal, inguinal, or perineal hernias.

Double-contrast cystography

This technique involves instillation of both positive and negative contrast media into the urinary bladder and is useful in evaluating mural and luminal bladder lesions.^{1,2,13,14,16,17,28,33–36} It is the method of choice for the evaluation of the urinary bladder if ultrasonography is not available.

Retrograde urethrography

Because of the at least partially intrapelvic location in dogs and cats and its length in male dogs, urethrography with instillation of a positive contrast medium into the urethra remains the imaging modality of choice to evaluate the entire organ.^{11,12,15,18,21}

The materials needed and techniques for these imaging studies are summarized in **Boxes 1–3**.

Box 1**Cystography and retrograde urethrography: materials**

- Catheter (type and size dependent on animal size [Foley catheter ideal])
- 3-way valve
- Sterile lubricant
- Syringes
- Sterile gloves
- Sterile saline
- ± Vaginal speculum for female dogs
- ± 2–5 mL of 2% lidocaine without epinephrine
- ± Catheter connectors
- Contrast medium:
 - Positive contrast cystogram:
 - Water-soluble iodinated contrast medium diluted with sterile saline to 15%–20% (150–200 mg iodine [I]/mL)
 - Dogs: 10 (3.5–13.1) mL/kg
 - Cats: 10–50 mL
 - Negative contrast cystogram:
 - Gaseous contrast agent (preferably CO₂ or N₂O)
 - Dose as above
 - Double-contrast cystogram:
 - Water-soluble iodinated contrast medium, undiluted
 - Dogs less than 25 lb: 1–3 mL
 - Dogs greater than 25 lb: 3–6 mL
 - Cats: 0.5–1 mL
 - Gaseous contrast agent (preferably CO₂ or N₂O)
 - Dogs: 10 (3.5–13.1) mL/kg
 - Cats: 10–50 mL
 - Retrograde urethrogram:
 - Water-soluble iodinated contrast medium diluted with sterile saline to 15%–20% (150–200 mg I/mL)
 - Dose highly variable dependent on patient size and lesion location
 - Contrast medium administered in 5–20 mL boluses each
 - Several repetitions may be needed for diagnosis

Box 2**Cystography: technique**

- Unless it is an emergency, assure adequate patient preparation
 - Fast patient
 - Administer enema if needed
- Sedation or anesthesia strongly recommended
 - Minimize patient pain/resistance
 - Optimize study quality
 - Reduce radiation exposure of examiner
- Survey lateral and ventrodorsal radiographs of the abdomen
 - Left lateral recumbency recommended by some to minimize risk of gas embolism
- Catheterization
 - Aseptic technique
 - Advance catheter to urinary bladder
 - Inflate cuff and retract catheter until the cuff is lodged within the trigone
 - Connect to 3-way valve
 - Empty bladder and, if needed, flush repeatedly with sterile saline to remove blood clots
- Administration of contrast medium
 - Important: To avoid overdistention of the bladder, injection should be performed under palpation control
 - Positive contrast cystography and negative contrast cystography:
 - Instill contrast agent until the bladder feels firm
 - Double-contrast cystography:
 - Instill negative contrast agent until bladder feels firm
 - Then instill positive contrast agent
 - Gently turn patient to assure even distribution of positive contrast medium along the bladder wall
- Obtain lateral, ventrodorsal, and, if needed, oblique radiographs

Other contrast procedures

Catheterization of the urethra in female animals may be challenging. If a catheter cannot be adequately positioned, a *vaginourethrography* can be performed during which the catheter tip is placed in the vaginal vestibule, an atraumatic clamp is placed on the vulva to avoid leakage, and contrast medium is injected to fill both the vagina and the urethra concurrently.^{12,37,38} *Antegrade* and *voiding cystography and urethrography* are technically challenging and not routinely performed. The reader is referred to other publications for further description of these techniques.^{4,39–41}

Normal findings

Assuming adequate distention, the normal urinary bladder has a wall thickness of 1 mm (cats) to 2 mm (dogs) and has a smooth mucosal surface and no discernible filling defects associated with wall or lumen.^{2,13,15,16} The position varies slightly with degree of distension during the contrast procedure.^{42,43} The normal urethra is

Box 3**Urethrography: technique**

- Unless it is in emergency, assure adequate patient preparation
- Sedation or anesthesia strongly recommended
- Survey lateral and ventrodorsal radiographs of the abdomen
- Catheterization
 - Aseptic technique
 - Initially advance catheter to urinary bladder
 - Connect to 3-way valve
 - Empty bladder and, if needed, flush repeatedly with sterile saline to remove blood clots
 - Distend urinary bladder with positive contrast medium (diluted with saline): this will ensure back pressure during subsequent injection of contrast medium into the urethra and adequate urethral distention, which will optimize evaluation of the wall. (However, normal structures such as the urethral crest may be obscured with this technique.)
 - Position catheter in the urethra
 - In cats and female dogs, the catheter tip should be located as distal in the urethra as possible
 - In male dogs, the evaluation of the entire urethra may require repositioning of the catheter and repeat injections, first from the ischiatic arch, the second from the caudal aspect of the os penis, and the third from the tip of the penis
 - If a balloon-tipped catheter is used, inflate balloon carefully
- Injection and radiographic documentation
 - Initial study most commonly performed in lateral recumbency
 - Fluoroscopy is optimal to monitor injection in real-time
 - If conventional radiographs are used, inject contrast bolus and obtain radiograph toward the end of the injection
 - Several injections may be needed dependent on initial findings
 - In male dogs, successively withdraw catheter and repeat injections from different urethral levels
 - If needed, repeat injections and obtain ventrodorsal and oblique radiographs

smoothly margined and may show longitudinal folds if incompletely distended. It should be of fairly even width with the following exceptions:

- The prostatic urethra in dogs is wider than the membranous urethra.⁴⁴
- In male dogs mild narrowing may be observed at the ischial arch.¹⁸
- The penile urethra in cats narrows progressively from the ischial arch to the external urethral orifice.¹⁵
- The colliculus seminalis may appear as a physiologic focal filling defect in male dogs and cats.^{4,28}
- In cats a physiologic filling defect is associated with the dorsal urethra representing the urethral crest.⁴⁵

Although an increase in urethral diameter with increased distension of the urinary bladder has been described in the proximal urethra in female dogs and cats and in the prostatic urethra of male dogs, a significant change was not observed in the

membranous and penile urethra in male dogs or in male cats.⁴⁶ Thin, linear reflux of contrast medium into the prostatic gland may be seen in some normal dogs and is not necessarily an indicator of disease.⁴⁷ Similarly, vesicoureteral reflux may be evident in some normal animals.⁴⁸

Ultrasonography

Technique and common artifacts

Patient preparation (clipping of the abdomen and application of alcohol and ultrasound gel) is performed in a routine manner.^{49,50} Instillation of a standardized amount of saline solution (5 mL/kg) has been advocated to allow more accurate measurements of bladder wall lesions especially if monitoring a patient's response to treatment of bladder neoplasia.⁵¹

An extensive discussion of ultrasound physics is beyond the scope of this article and the reader is referred to other references.^{52,53} However, a short overview of pertinent ultrasound artifacts is provided because these may be mistaken for lesions or may obscure the area of interest. Artifacts commonly encountered when imaging the urinary tract include acoustic shadowing, reverberation artifact, distal acoustic enhancement, slice thickness artifact, grating or side-lobe artifact, edge shadowing (refraction) artifact, and twinkle artifact.^{54–60}

Acoustic shadowing occurs because of mineral or air interfaces causing loss of echo intensity deep to the interface because of absorption and/or reflection. Mineral interfaces such as uroliths typically cause strong distal hypoechoic to anechoic shadows. Air interfaces typically cause echogenic ("dirty") distal shadows because of concurrent reverberation artifact. This artifact is seen when gas is present within the bladder lumen or wall.⁵⁷

Distal acoustic enhancement is due to passage of sound through an area of decreased attenuation resulting in higher echo intensity from tissues deep to the area compared with surrounding tissues at the same depth. This artifact is typically seen with fluid-filled structures such as the urinary bladder.⁵⁷

Slice thickness, side-lobe, and grating-lobe artifacts may cause the appearance of echoic material within anechoic areas. These artifacts can give the false impression of sediment within the urinary bladder.⁵⁴

Edge shadowing artifact is caused by refraction of the ultrasound beam. It occurs at the margin of curved structures and results in a dark line or shadow extending distally. If this artifact involves a curved fluid-filled structure (bladder) surrounded by fluid (abdominal effusion), it will cause the impression of a defect in the bladder wall (**Fig. 3**).⁵⁶

Twinkling artifact is a color Doppler artifact that occurs behind strongly reflective interfaces such as those produced by urinary tract stones or parenchymal calcifications. It appears as a quickly fluctuating mixture of Doppler signals with an associated spectrum of noisy appearance and can be used to differentiate mineral structures from other hyperechoic material.⁵⁸

Normal findings

The urinary bladder is found in the caudoventral abdomen and can vary greatly in normal size because of distention with urine.^{59,60} The colon lies dorsally or dorsolaterally and often indents the bladder wall when the bladder is not fully distended. Urinary bladder wall thickness should be measured when the bladder is adequately distended; wall thickness measurements of an empty bladder are significantly thicker and are not reliable.⁶¹ Normal wall thickness should be less than 2 mm in normal dogs, with larger dogs having a slightly thicker wall than smaller dogs. Normal bladder wall thickness varies from 1.3 to 1.7 mm in cats.⁶² Lower-frequency transducers show

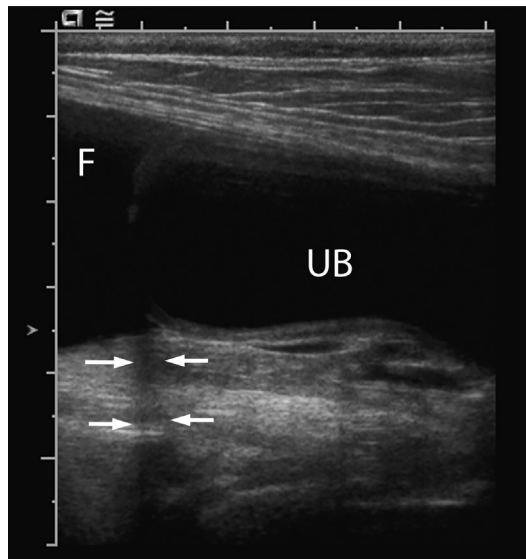


Fig. 3. Edge shadowing artifact in a dog with abdominal effusion (F). There is refraction of the ultrasound beam at the cranial (*curved*) margin of the urinary bladder (UB), resulting in the impression of a bladder wall defect. A dark line extending distally is also noted (*arrows*).

the bladder wall as a single echogenic line; however, high-frequency transducers typically show distinct bladder wall layers.⁵⁰ Normal ureteral papillae may be seen in some animals as small focal protrusions of the dorsal wall and should not be confused with abnormal wall thickening or bladder masses.⁵⁵ Ureteral “jetting” of urine may on occasion be observed and helps identifying the ureteral openings. Position of the bladder neck within the pelvic canal in some dogs makes complete evaluation difficult because of overlying bone.

Normal dog and cat urine is anechoic. Echogenic debris seen in the urine may be due to crystals, protein, cells, cellular debris, calculi, or fat droplets, and urinalysis is necessary for evaluation of the clinical significance.^{49,59,60}

The urethra extends from an ill-defined juncture with the bladder neck into the pelvic canal as a hypoechoic tubular structure. Normally the lumen of the urethra does not contain observable fluid. The intrapelvic urethra is not usually visualized because of overlying pelvic bone but may be observable through the obturator foramen in some cases.^{50,59,60} Transrectal imaging of the urethra can be performed but requires expensive specialized transducers.⁶³ In male dogs part of the prostatic and penile urethra may be evaluated from a perineal and/or ventral approach. Although the mucosal margin of the urethra is best assessed by positive contrast urethrography, ultrasound offers complementary information and can be used to assess urethral thickness and echogenicity.⁶⁴

Advanced Diagnostic Imaging (Computed Tomography, Magnetic Resonance Imaging)

With increasing availability, the use of cross-sectional imaging modalities (computed tomography [CT]; magnetic resonance imaging [MRI]) for diagnostic imaging of the abdomen including the lower urinary tract in small animals is likely to increase. Initial applications described include the use of CT excretory urography to evaluate the

ureterovesicular junction in patients with suspected ureteral ectopia and the use of either CT or MRI to evaluate intrapelvic lesions.^{65–68} Both CT and MRI allow excellent anatomic delineation and are useful in assessing origin and extent of intrapelvic masses, identifying lesions associated with the pelvic urethra such as wall thickening or calculi, and evaluating surrounding structures including the spine and regional lymph nodes for the presence of additional lesions (eg, metastatic disease). CT has proven superior to ultrasound in accurately measuring the size of bladder wall masses.⁵¹ CT and MRI are useful in planning a surgical approach for intrapelvic lesions and may be needed if radiation therapy planning is to be pursued for pelvic canal neoplasia. Finally, 3-dimensional (3D) volume rendering and virtual endoscopy using CT and MRI datasets are increasingly used in humans and are likely to gain popularity in veterinary medicine with increasing availability of these techniques.^{69–74} A detailed discussion of CT and MRI principles, physics, and applications is beyond the scope of this article and readers are referred to specific references.⁷⁵ Techniques that have been suggested for evaluation of the ureterovesicular junction^{66,68} and volume measurements of bladder masses in dogs,⁵¹ respectively, are shown in **Boxes 4** and **5**.

Normal findings

The urinary bladder is located within the caudal abdomen and is filled with urine of homogenous attenuation/intensity unless contrast medium has been administered, which

Box 4

Evaluation of the ureterovesicular junction using CT-excretory urography: technique

- Fast patient for 12 hours and allow to urinate/defecate 2–4 hours before procedure (± enema)
- Place intravenous catheter in peripheral vein
- Sedation or general anesthesia
- (± Place urinary catheter, empty bladder, and distend bladder with negative contrast medium to improve delineation between urinary bladder and ureters)
- Position patient in sternal recumbency
- Precontrast scan from cranial level of kidneys to tuber ischii (± second scan of pelvic canal with smaller field of view)
 - Tube factors: 120 kVp and 120–250 mA
 - Slice thickness: 3–5 mm (thinner if possible dependent on equipment)
- Manual IV injection: Iodinated contrast medium, 400 mg I/kg
- Postcontrast scan (either dynamic or conventional):
 - Dynamic acquisition
 - Slice position stationary and caudal bladder neck
 - 5–10 minutes after injection: Obtain slices at 1-second slice intervals for 30 seconds
 - Standard acquisition:
 - Repeat precontrast scan(s) 3 minutes after injection
 - (± delayed scan [5–10 minutes] if ureters not adequately visualized during first scan)

Modified from Rozear L, Tidwell AS. Evaluation of the ureter and ureterovesicular junction using helical computed tomographic excretory urography in healthy dogs. *Vet Radiol Ultrasound* 2003;44(2):155–64; and Schwarz T. Urinary system. In: Schwarz T, Saunders J, editors. *Veterinary computed tomography*. Chichester (United Kingdom): John Wiley & Sons Ltd; 2011. p. 331–8.

Box 5**Urinary bladder CT protocol for tumor volume measurements**

- Place IV catheter and induce anesthesia
- Place of an indwelling urinary catheter
- Position patient in dorsal recumbency
- Remove urine from urinary bladder and insufflate CO₂ (5 mL/kg)
- Scan caudal abdomen (1.25-mm slice thickness) from caudal margin of kidneys to pelvic urethra immediately following IV administration of iodinated contrast medium (600 mg I/kg)

From Naughton JE, Widmer WR, Constable PD, et al. Accuracy of three-dimensional and two-dimensional ultrasonography for measurement of tumor volume in dogs with transitional cell carcinoma of the urinary bladder. Am J Vet Res 2012;73(12):1919–24.

will result in a layered appearance.⁶⁵ Typically, contrast medium accumulates in the dependent portion of the urinary bladder; however, an inverted layering has been reported in CT and has been attributed to increased urine specific gravity.⁷⁶ Similar as described for contrast studies and ultrasonography, normal wall thickness of an adequately distended urinary bladder should range from 1 to 2 mm. If intravenous contrast medium has been administered, the ureters are seen entering the urinary bladder at the level of the trigone (**Fig. 4**).^{66,68} The normal urethra can be traced caudally from the bladder neck and is clearly delineated within the pelvic canal as a homogenous tubular structure, which should not contain a significant amount of fluid.⁶⁵

IMAGING OF SPECIFIC LOWER URINARY TRACT DISORDERS

Cystitis and Urethritis

Cystitis is common in dogs and cats. Imaging may not show any abnormalities in early or mild cases. In chronic and more severe cases abdominal ultrasound or double-contrast cystography may show diffuse thickening and irregularity of the urinary bladder wall, especially in the cranioventral portion of the bladder (**Fig. 5**, see also **Fig. 10**).^{36,77,78} Additional findings might include intraluminal material due to the



Fig. 4. CT-Excretory urography of the normal ureterovesical junction in a 3-year-old female spayed Swiss mountain dog. The left ureteral opening is associated with the dorsal bladder wall. Positive contrast medium is seen entering the bladder and accumulating ventrally. The right ureteral opening was located slightly more cranially and is not visible on this image.

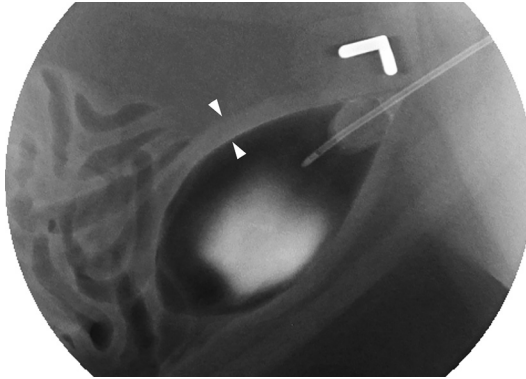


Fig. 5. Double-contrast cystography showing severe generalized bladder wall thickening (between arrowheads) in a 5-year-old male castrated domestic shorthair cat with chronic cystitis.

presence of inflammatory products and/or hemorrhage, mineral sediment, or cystic calculi. Mineralization of the bladder wall secondary to severe chronic cystitis has been described but is very rare.⁷⁹

Emphysematous cystitis due to infection with gas-producing bacteria such as *Escherichia coli* or *Clostridium* species is most commonly seen in animals with diabetes mellitus but may affect other patients as well. A diagnosis is usually possible on survey radiographs where variable size gas opacities are associated with the bladder wall (and possibly the lumen) (Fig. 6).^{6,80,81} Ultrasonographically, these gas inclusions appear as hyperechoic areas with distal reverberation artifacts.^{82,83} Care must be taken to distinguish these pathologic gas accumulations from intraluminal gas introduced during cystocentesis, catheterization, or endoscopy, and this can be accomplished by repositioning the dog. While free luminal gas will move with a change in patient position, gas within the wall will remain static.⁵⁹

Polypoid cystitis is a rare disease of the urinary bladder in dogs characterized by inflammation, epithelial proliferation, and development of a polypoid to pedunculated mass or masses without histopathologic evidence of neoplasia.^{84,85} Concurrent cystic calculi are common. Most of the masses are located cranioventrally in the bladder (Fig. 7) as opposed to transitional cell carcinoma (TCC), which has a predilection for



Fig. 6. Emphysematous cystitis in a 12-year-old female spayed Labrador retriever. Linear gas inclusions are associated with the ventral bladder wall (arrow). Small bladder calculi are also present.

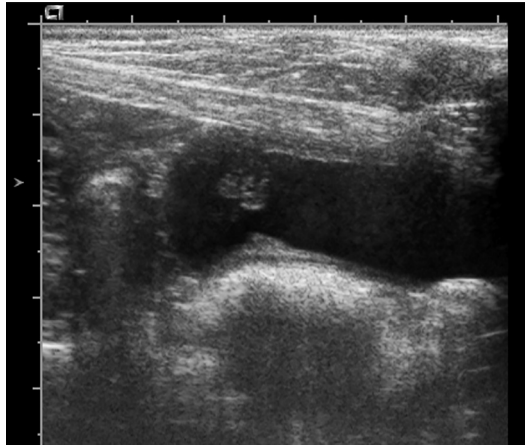


Fig. 7. Polypoid cystitis in a 5-year-old male castrated shih tzu. The ultrasonographic image shows a pedunculated mass attached to the cranial ventral bladder wall by a thin stalk.

the bladder neck or trigone area. However, because bladder neoplasia cannot be excluded based on cystographic or ultrasonographic findings, biopsy may be warranted to establish a definitive diagnosis.

Urethritis is uncommon in small animals and is usually associated with cystitis, prostatitis, or vaginitis.⁸⁶ Abdominal ultrasound or a urethrogram shows diffuse and irregular thickening of the urethra, which in the case of granulomatous urethritis may mimic urethral neoplasia.⁸⁷

Urolithiasis

Bladder and urethral calculi are common. In the case of radiopaque calculi (calcium oxalate, calcium phosphate, and struvite) a diagnosis is usually easily possible on survey radiographs where they appear as rounded to spiculated mineral opacities of variable size in the plane of the urinary bladder and/or urethra (**Fig. 8**, see also **Figs. 1** and **2**).² Even though contrast resolution of modern digital radiography equipment seems to have improved detectability of less opaque and nonopaque calculi (silica, urate, and cystine calculi) compared with previous film screen radiography (**Fig. 9**),



Fig. 8. Lateral survey radiograph of the caudal abdomen showing numerous large well-circumscribed bladder calculi in a 9-year-old female spayed Staffordshire terrier.



Fig. 9. Lateral radiograph of the caudal abdomen in a 5-year-old male Dalmatian (digital radiograph). Small intestinal segments superimposed over the urinary bladder interfere with evaluation. There are numerous minimally opaque bladder calculi, consistent with urate urolithiasis.

other imaging procedures may be needed to identify calculi, especially if obscured by overlying pelvic bones.^{7,88} Using abdominal ultrasound, bladder stones appear as single or multiple strongly hyperechoic structures of variable size and shape (**Fig. 10**).⁸⁹ Dependent on size of the calculus and transducer frequency, distal shadowing may be present.⁹⁰ Sensitivity of ultrasonographic examination for the detection of cystic calculi is high and similar to double-contrast cystography if using a high-frequency (7.5 MHz or above) transducer. However, if a lower-frequency (5 MHz or lower) transducer is used, false negative results are common especially with small stones.⁷ Although size measurements of cystoliths using radiography, double-contrast cystography, and CT are comparable and considered accurate, ultrasonographic measurements may overestimate the true size of calculi and should be interpreted with caution whenever cystolith size may influence patient management.⁹¹ Possible concurrent findings in animals with bladder stones include bladder wall thickening, uroliths

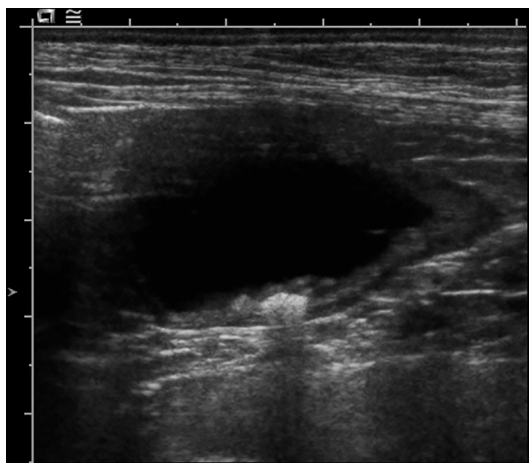


Fig. 10. Ultrasound image of the urinary bladder in an 11-year-old male shih tzu showing bladder calculi with moderate distal acoustic shadowing and mild bladder wall thickening consistent with cystitis.

associated with kidneys, ureters, or urethra, and inflammatory renal changes (pyelectasia, hyperechogenicity of renal cortices, and others).

Urinary calculi may become lodged in the urethra and result in partial or complete urethral obstruction. If the calculus is located in the intra-abdominal portion of the urethra or in the penile urethra in male dogs, it can usually be visualized by means of ultrasonography. Distension of the visible portion of the urethra and failure to visualize an obstructive lesion necessitate further imaging (positive contrast urethrography or possibly CT) (**Fig. 11**).

Neoplasms of the Urinary Bladder and Urethra

Urinary bladder tumors are most commonly identified by means of abdominal ultrasonography; alternatively, double-contrast cystography can be used. Survey radiographs are usually not helpful in identifying the primary lesion, although tumoral dystrophic mineralization may on occasion be visible in the periphery of the urinary bladder.¹⁴ Radiographs are, however, useful in evaluating the patient for metastatic disease to sublumbar lymph nodes, pelvis, and spine. Transitional cell carcinoma (TCC) is the most common tumor type.⁹² Other epithelial tumors include transitional cell papillomas, squamous cell carcinomas, adenocarcinomas, and undifferentiated carcinomas. Mesenchymal tumors are less common and include fibromas, fibrosarcomas, leiomyomas, leiomyosarcomas, rhabdomyosarcomas, lymphosarcomas, hemangiomas, and hemangiosarcomas.⁹³ TCCs are predominantly located at the level of the trigone but may be found in any location. Diffuse infiltration of the bladder wall instead of formation of a distinct mass is also possible.⁵⁹ Double-contrast cystography typically demonstrates one or more broad-based irregularly margined wall-associated filling defects.^{1,4,14,34} On abdominal ultrasound bladder TCCs are usually sessile, irregularly shaped masses of variable echogenicity (**Fig. 12**).^{59,60,77,94} Concurrent findings may include hydronephrosis/hydroureter secondary to ureteral obstruction and evidence of metastatic disease to the medial iliac lymph nodes or other abdominal organs. Ultrasonographic findings of wall involvement, heterogeneous mass, and trigone location are associated with significantly shorter survival times and may be used as prognostic indicators.⁹⁵ Conventional 2-dimensional (2D) ultrasound has proven inferior to 3D ultrasound and cross-sectional imaging in obtaining reliable tumor volume measurements.⁵¹ However, these modalities are currently



Fig. 11. Urethrogram in a male dog demonstrating a sharply margined intraluminal filling defect (*arrow*), consistent with a urethral calculus.

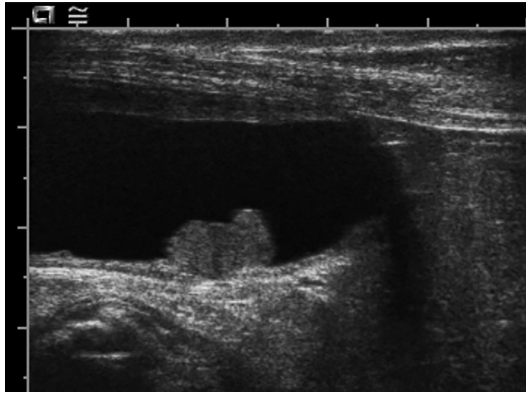


Fig. 12. TCC of the urinary bladder in an 8-year-old female spayed beagle. The sagittal ultrasonographic image demonstrates an approximately 1-cm-length broad-based lobulated mass associated with the dorsal bladder wall. The ventral bladder wall also appears thickened.

limited to specialty institutions. When monitoring a patient's response to treatment using conventional 2D ultrasound, it is important to keep the ultrasound technique and protocol (eg, positioning, degree of bladder filling, radiologist) as consistent as possible between examinations to minimize the chance of measurement errors. Smooth muscle tumors of the urinary bladder appear as smoothly margined hypoechoic or heterogeneous masses.^{59,96} Lymphoma of the urinary bladder wall has been reported to cause masses of variable echogenicity, size, and location.⁹⁷ Concurrent ureteral obstruction and lymphadenopathy are common. Differentiation from other tumor types is not possible by means of ultrasonography.

Most urethral tumors are malignant epithelial tumors (TCC or squamous cell carcinoma). A presumptive diagnosis is most commonly established by means of abdominal ultrasound or contrast urethrography. Urethrography demonstrates thickening of the urethra and marked irregularity to the mucosal lining resulting in a moth-eaten appearance (**Fig. 13**).⁹⁸ Ultrasound demonstrates hypoechogenicity and irregular thickening of the urethra, with a hyperechoic line along the epithelial surface.⁶⁴



Fig. 13. Urethral TCC in a 10-year-old female spayed German shepherd dog. The urethrograph shows that the urethral wall is severely generally thickened, with marked irregularity of the mucosal surface.

Possible concurrent findings include abnormalities of bladder and/or prostatic gland, hydronephrosis, and medial iliac lymph node enlargement.

Imaging findings in other urinary bladder and urethral tumors are rarely reported and variable.^{59,99–101} Neoplasms of structures bordering the urethra, such as prostate, os penis, or vagina/uterus, may involve the urethra, and origin of the disease process may not be obvious in some cases.^{102–104}

Advanced diagnostic imaging (CT/MRI) of the bladder is at this point not commonly performed in small animal patients with bladder and urethral neoplasms because of associated cost, need for heavy sedation/general anesthesia, and lack of availability. Both CT and MRI are excellent modalities in evaluating the primary lesion (**Fig. 14**) as well as in the extent of metastatic disease to lymph nodes and osseous structures (**Fig. 15**).^{65,68} CT cystography (see **Box 5**) provides an excellent method for accurate tumor volume measurements and should be considered especially if evaluating a patient for possible surgery or when monitoring a patient's response to treatment.⁵¹ In addition, virtual endoscopy using CT or MRI data sets (**Fig. 16**) is increasingly used in human medicine and is likely to gain popularity in veterinary medicine with increasing availability.^{70–74}

Because there is overlap in the imaging appearance of various bladder/urethral tumors as well as tumors and inflammatory conditions (eg, polypoid cystitis or granulomatous urethritis), biopsy is necessary for a definitive diagnosis (**Fig. 17**). Samples of



Fig. 14. Bladder TCC in a 6-year-old female spayed West Highland white terrier. Dorsal multiplanar reconstructed CT image obtained after instillation of positive contrast medium into the urinary bladder demonstrates a large broad-based irregularly margined filling defect (mass) associated with the left bladder wall.

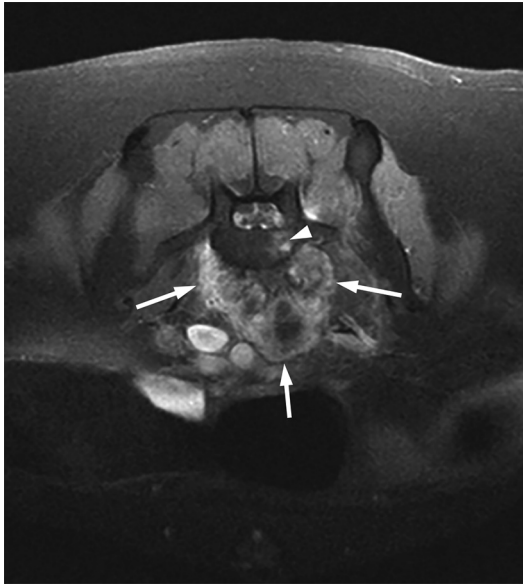


Fig. 15. Metastatic disease from urethral and/or prostatic carcinoma in a 10-year-old male castrated otterhound. The postcontrast T1-weighted fat-saturated image shows a large sub-lumbar mass abutting the ventral lumbar vertebral bodies (*arrows*), focal contrast enhancement within the left aspect of the vertebral body (*arrowhead*), mild contrast enhancement of the left epaxial muscles, and extradural compressive contrast-enhancing material within the ventral aspect of the vertebral canal.

bladder or urethral masses should be obtained by means of ultrasound-guided traumatic catheterization (“suction biopsy”) instead of percutaneous tissue sampling due to the risk of tumor cell implantation along the needle tract¹⁰⁵; this is especially important for apical tumors in which surgery may be possible.⁸

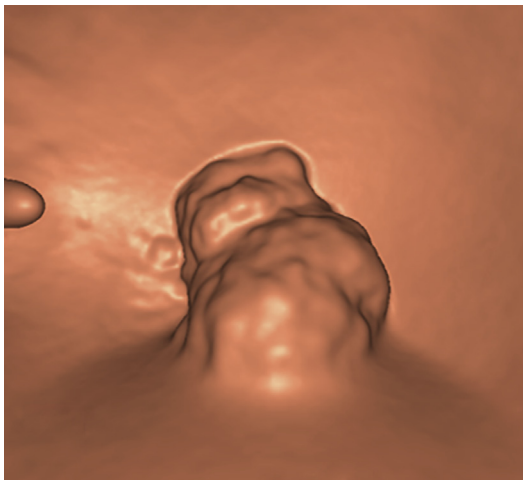


Fig. 16. Virtual endoscopic image generated from a CT data set in a dog with TCC of the urinary bladder (same patient as **Fig. 14**). The 3D display of the mass is comparable to what is seen when performing cystoscopy.

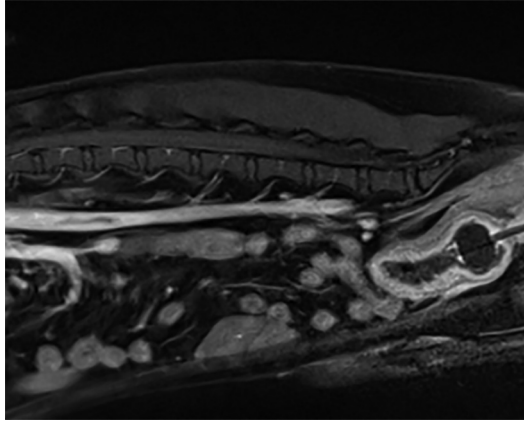


Fig. 17. Severe infiltrative bladder disease in an 8-year-old male castrated bichon frise presented with back pain and urethral obstruction. Prior fine-needle aspiration cytology of the markedly thickened bladder wall yielded a diagnosis of "marked neutrophilic inflammation and possible carcinoma." The postcontrast T1-weighted fat-saturated image shows that the urinary bladder wall is severely thickened and has a markedly irregular mucosal surface. An indwelling Foley catheter is present within the bladder and urethra. There is no evidence of metastatic disease to the spine or sublumbar lymph nodes. Subsequent surgical biopsies yielded the diagnosis of "marked necroulcerative cystitis."

Urinary Bladder and Urethral Rupture

Although the use of contrast cystosonography has been described as a sensitive tool for bladder rupture, positive contrast cystography provides more information on location of the rupture than ultrasound. Similarly, urethrography is considered superior to ultrasonography for the diagnosis of urethral rupture and will show extravasation of contrast medium into surrounding soft tissues and/or peritoneal cavity (**Fig. 18**).^{106,107}



Fig. 18. Urethral rupture in a 1-year-old male miniature dachshund presented after being hit by car. The urethrogram demonstrates extravasation of contrast medium from the urethra into the perineal soft tissues and caudoventral peritoneal cavity.

Other

Abnormal position of the urinary bladder may be observed secondary to displacement by adjacent organs or masses or secondary to an abdominal, inguinal, or perineal hernia. Although ultrasound can be helpful in identifying an abnormally positioned bladder, positive contrast cystography may be given preference especially if the bladder is partially obscured by the pelvis.¹³ A focal outpouching from the cranial ventral bladder margin seen on ultrasound or cystogram is consistent with a *persistent urachus/urachal diverticulum*, which is an occasional incidental finding or may be associated with chronic/recurrent cystitis.^{2,108,109} An *abnormal communication* (fistula) between the bladder or urethra and other structures (eg, rectum) may be observed as a congenital abnormality or secondary to trauma.^{30–32,110–112}; this is best diagnosed by means of positive contrast cystography or urethrography. Although *cystic foreign* bodies in humans are not uncommon and are usually self-inflicted,¹¹³ they are very rare in animals. Potential causes include migration of foreign material from other organs, ballistic fragments, and urinary catheter pieces after catheter failure.^{114–116}

SUMMARY

Lower urinary tract disorders are common in small animal patients, and diagnostic imaging is routinely performed as part of the diagnostic workup. Survey radiographs of the abdomen are helpful in identifying radiopaque bladder or urethral calculi, gas within the urinary tract, and enlarged lymph nodes or bone metastases secondary to lower urinary tract neoplasia. Contrast procedures (cystography and urethrography) have largely been replaced with ultrasonography but remain useful in the evaluation of bladder or urethral rupture, abnormal communication with other organs, and lesions of the pelvic or penile urethra. Ultrasonography is the modality of choice for the diagnosis of most disorders affecting the bladder and intra-abdominal portion of the urethra. Advanced diagnostic imaging modalities (CT, MRI) are useful in assessing the ureterovesical junction, evaluating intrapelvic lesions, improving surgical planning, monitoring the size of known lesions associated with the urinary tract, and evaluating regional lymph nodes and osseous structures for metastases.

REFERENCES

1. Johnston GR, Feeney DA. Comparative organ imaging—lower urinary tract. *Vet Radiol* 1984;25(4):146–53.
2. Marolf AJ, Park RD. The urinary bladder. In: Thrall DE, editor. *Textbook of veterinary diagnostic radiology*. 6th edition. St Louis (MO): Elsevier Saunders; 2013. p. 726–43.
3. Johnston GR, Feeney DA, Rivers WJ, et al. Diagnostic imaging of the feline lower urinary tract. *Vet Clin North Am Small Anim Pract* 1996;26(2):401–15.
4. Ackerman N. Radiology and ultrasound of urogenital diseases in dogs and cats. Ames (IA): Iowa State University Press; 1991.
5. Norris AM, Laing EJ, Valli VE, et al. Canine bladder and urethral tumors: a retrospective study of 115 cases (1980–1985). *J Vet Intern Med* 1992;6(3):145–53.
6. Root CR, Scott RC. Emphysematous cystitis and other radiographic manifestations of diabetes mellitus in dogs and cats. *J Am Vet Med Assoc* 1971;158(6):721–8.
7. Weichselbaum RC, Feeney DA, Jessen CR, et al. Urocystolith detection: comparison of survey, contrast radiographic and ultrasonographic techniques in an in vitro bladder phantom. *Vet Radiol Ultrasound* 1999;40(4):386–400.

8. Allstadt SD, Lee N, Scruggs JL, et al. Clinical rounds—transitional cell carcinoma. *Vet Med* 2014;109(10):327–33.
9. Adams WM, Dibartola SP. Radiographic and clinical features of pelvic bladder in the dog. *J Am Vet Med Assoc* 1983;182(11):1212–7.
10. Mahaffey MB, Barsanti JA, Barber DL, et al. Pelvic bladder in dogs without urinary incontinence. *J Am Vet Med Assoc* 1984;184(12):1477–9.
11. Ackerman N. Positive-contrast retrograde urethrography. *Mod Vet Pract* 1980; 61(8):684–6.
12. Brown JC. The urethra. In: Thrall DE, editor. *Textbook of veterinary diagnostic radiology*. 6th edition. St Louis (MO): Elsevier Saunders; 2013. p. 744–56.
13. Essman SC. Contrast cystography. *Clin Tech Small Anim Pract* 2005;20(1):46–51.
14. Feeney DA, Anderson KL. Radiographic imaging in urinary tract disease. In: Bartges J, Polzin DJ, editors. *Nephrology and urology of small animals*. Ames (IA): Wiley-Blackwell; 2011. p. 97–127.
15. Johnston GR, Feeney DA, Osborne CA. Urethrography and cystography in cats .1. Techniques, normal radiographic anatomy, and artifacts. *Comp Cont Ed Pract Vet* 1982;4(10):823–36.
16. Osborne CA, Jessen CR. Double-contrast cystography in the dog. *J Am Vet Med Assoc* 1971;159(11):1400–4.
17. Pugh CR, Rhodes WH, Biery DN. Contrast studies of the urogenital system. *Vet Clin North Am Small Anim Pract* 1993;23(2):281–306.
18. Ticer JW, Spencer CP, Ackerman N. Positive contrast retrograde urethrography—a useful procedure for evaluating urethral disorders in the dog. *Vet Radiol* 1980;21(1):2–11.
19. Barsanti JA, Crowell W, Losonsky J, et al. Complications of bladder distension during retrograde urethrography. *Am J Vet Res* 1981;42(5):819–21.
20. Farrow CS. Exercise in diagnostic radiology: intimal dissection and luminal involution secondary to double contrast cystography. *Can Vet J* 1981;22(8):260–1.
21. Johnston GR, Feeney DA, Osborne CA. Urethrography and cystography in cats .2. Abnormal radiographic anatomy and complications. *Comp Cont Ed Pract Vet* 1982;4(4):931–46.
22. Johnston GR, Stevens JB, Jessen CR, et al. Complications of retrograde contrast urethrography in dogs and cats. *Am J Vet Res* 1983;44(7):1248–56.
23. Ackerman N, Wingfield WE, Corley EA. Fatal air embolism associated with pneumourethrography and pneumocystography in a dog. *J Am Vet Med Assoc* 1972; 160(12):1616–8.
24. Zontine WJ, Andrews LK. Fatal air embolization as a complication of pneumocystography in two cats. *J Am Vet Rad Soc* 1978;19(1):8–11.
25. Breton L, Pennock PW, Valli VE. Effects of hypaque 25-percent and sodium iodide 10-percent in canine urinary bladder. *J Am Vet Rad Soc* 1978;19(4): 116–24.
26. Feeney DA, Osborne CA, Jessen CR. Effects of radiographic contrast-media on results of urinalysis, with emphasis on alteration in specific gravity. *J Am Vet Med Assoc* 1980;176(12):1378–81.
27. Ruby AL, Ling GV, Ackerman N. Effect of sodium diatrizoate on the in vitro growth of three common canine urinary bacterial species. *Vet Radiol* 1983; 24(5):222–5.
28. Feeney DA, Johnston GR. Urogenital imaging: a practical update. *Semin Vet Med Surg (Small Anim)* 1986;1(2):144–64.
29. Silverman S, Long CD. The diagnosis of urinary incontinence and abnormal urination in dogs and cats. *Vet Clin North Am Small Anim Pract* 2000;30(2):427–48.

30. Silverstone AM, Adams WM. Radiographic diagnosis of a rectourethral fistula in a dog. *J Am Anim Hosp Assoc* 2001;37(6):573–6.
31. Osuna DJ, Stone EA, Metcalf MR. A urethrorectal fistula with concurrent urolithiasis in a dog. *J Am Anim Hosp Assoc* 1989;25(1):35–9.
32. Waknitz D, Greer DH. Urethrorectal fistula in a cat. *Vet Med Small Anim Clinician* 1983;78(1):1551–3.
33. Bischoff MG. Radiographic techniques and interpretation of the acute abdomen. *Clin Tech Small Anim Pract* 2003;18(1):7–19.
34. Scrivani PV, Léveillé R, Collins RL. The effect of patient positioning on mural filling defects during double contrast cystography. *Vet Radiol Ultrasound* 1997;38(5):355–9.
35. Mahaffey MB, Barber DL, Barsanti JA, et al. Simultaneous double-contrast cystography and cystometry in dogs. *Vet Radiol* 1984;25(6):254–9.
36. Mahaffey MB, Barsanti JA, Crowell WA, et al. Cystography—effect of technique on diagnosis of cystitis in dogs. *Vet Radiol* 1989;30(6):261–7.
37. Léveillé R, Atilola MA. Retrograde vaginocystography—a contrast study for evaluation of bitches with urinary incontinence. *Compend Contin Educ Vet* 1991;13(6):934–41.
38. Rivers B, Johnston GR. Diagnostic imaging of the reproductive organs of the bitch—methods and limitations. *Vet Clin North Am Small Anim Pract* 1991;21(3):437–66.
39. Moreau PM, Lees GE, Gross DR. Simultaneous cystometry and uroflowmetry (micturition study) for evaluation of the caudal part of the urinary tract in dogs—studies of the technique. *Am J Vet Res* 1983;44(9):1769–73.
40. Moreau PM, Lees GE, Gross DR. Simultaneous cystometry and uroflowmetry (micturition study) for evaluation of the caudal part of the urinary tract in dogs: reference values for healthy animals sedated with xylazine. *Am J Vet Res* 1983;44(9):1774–81.
41. Poogird W, Wood AK. Radiologic study of the canine urethra. *Am J Vet Res* 1986;47(12):2491–7.
42. Johnston GR, Osborne CA, Jessen CR. Effects of urinary bladder distension on the length of the dog and cat urethra. *Am J Vet Res* 1985;46(2):509–12.
43. Johnston GR, Osborne CA, Jessen CR, et al. Effects of urinary bladder distension on location of the urinary bladder and urethra of healthy dogs and cats. *Am J Vet Res* 1986;47(2):404–15.
44. Feeney DA, Johnston GR, Osborne CA, et al. Dimensions of the prostatic and membranous urethra in normal male dogs during maximum distension retrograde urethrocytography. *Vet Radiol* 1984;25(6):249–53.
45. Scrivani PV, Chew DJ, Buffington CA, et al. Results of retrograde urethrography in cats with idiopathic, nonobstructive lower urinary tract disease and their association with pathogenesis: 53 cases (1993–1995). *J Am Vet Med Assoc* 1997;211(6):741–8.
46. Johnston GR, Jessen CR, Osborne CA. Effects of bladder distension on canine and feline retrograde urethrography. *Vet Radiol* 1983;24(6):271–7.
47. Ackerman N. Prostatic reflux during positive contrast retrograde urethrography in the dog. *Vet Radiol* 1983;24(6):251–9.
48. Christie BA. Vesicoureteral reflux in dogs. *J Am Vet Med Assoc* 1973;162(9):772–6.
49. Hecht S. Applications of ultrasound in diagnosis and management of urinary disease. In: Bonagura J, Twedt D, editors. *Kirk's current veterinary therapy XV*. St Louis (MO): Elsevier Saunders; 2014. p. 840–5.

50. Hecht S, Henry GA. Ultrasonography of the urinary tract. In: Bartges J, Polzin DJ, editors. Nephrology and urology of small animals. Ames (IA): Wiley-Blackwell; 2011. p. 128–45.
51. Naughton JE, Widmer WR, Constable PD, et al. Accuracy of three-dimensional and two-dimensional ultrasonography for measurement of tumor volume in dogs with transitional cell carcinoma of the urinary bladder. *Am J Vet Res* 2012;73(12): 1919–24.
52. Kremkau FW. Doppler ultrasound—principles and instruments. 2nd edition. Philadelphia: W. B. Saunders Company; 1995.
53. Kremkau FW. Diagnostic ultrasound. 6th edition. Philadelphia: W. B. Saunders Company; 2002.
54. Barthez PY, Léveillé R, Scrivani PV. Side lobes and grating lobes artifacts in ultrasound imaging. *Vet Radiol Ultrasound* 1997;38(5):387–93.
55. Douglass JP. Bladder wall mass effect caused by the intramural portion of the canine ureter. *Vet Radiol Ultrasound* 1993;34(2):107.
56. Douglass JP, Kremkau FW. Ultrasound corner—the urinary bladder wall hypoechoic pseudolesion. *Vet Radiol Ultrasound* 1993;34(1):45–6.
57. Kirberger RM. Imaging artifacts in diagnostic ultrasound—a review. *Vet Radiol Ultrasound* 1995;36(4):297–306.
58. Louvet A. Twinkling artifact in small animal color-Doppler sonography. *Vet Radiol Ultrasound* 2006;47(4):384–90.
59. Sutherland-Smith J. Bladder and urethra. In: Penninck D, d'Anjou MA, editors. Atlas of small animal ultrasonography. Ames (IA): Blackwell Publishing; 2008. p. 365–83.
60. Barrett E. Bladder and urethra. In: Barr F, Gaschen L, editors. BSAVA manual of canine and feline ultrasonography. Gloucester (United Kingdom): British Small Animal Veterinary Association; 2011. p. 155–64.
61. Geisse AL, Lowry JE, Schaeffer DJ, et al. Sonographic evaluation of urinary bladder wall thickness in normal dogs. *Vet Radiol Ultrasound* 1997;38(2):132–7.
62. Finn-Bodner ST. The urinary bladder. In: Cartee RE, Selcer BA, Hudson JA, et al, editors. Practical veterinary ultrasound. Philadelphia: Lea and Febiger; 1995. p. 210–35.
63. Levy DA, Cromeens DM, Evans R, et al. Transrectal ultrasound-guided intraprostatic injection of absolute ethanol with and without carmustine: a feasibility study in the canine model. *Urology* 1999;53(6):1245–51.
64. Hanson JA, Tidwell AS. Ultrasonographic appearance of urethral transitional cell carcinoma in ten dogs. *Vet Radiol Ultrasound* 1996;37(4):293–9.
65. MacLeod AG, Wisner ER. Computed tomography and magnetic resonance imaging of the urinary tract. In: Bartges J, Polzin DJ, editors. Nephrology and urology of small animals. Ames (IA): Wiley-Blackwell; 2011. p. 146–60.
66. Rozear L, Tidwell AS. Evaluation of the ureter and ureterovesicular junction using helical computed tomographic excretory urography in healthy dogs. *Vet Radiol Ultrasound* 2003;44(2):155–64.
67. Samii VF, McLoughlin MA, Mattoon JS, et al. Digital fluoroscopic excretory urography, digital fluoroscopic urethrography, helical computed tomography, and cystoscopy in 24 dogs with suspected ureteral ectopia. *J Vet Intern Med* 2004;18(3):271–81.
68. Schwarz T. Urinary system. In: Schwarz T, Saunders J, editors. Veterinary computed tomography. Chichester (United Kingdom): John Wiley & Sons Ltd; 2011. p. 331–8.
69. Orabi H, Aboushwareb T, Tan J, et al. Can computed tomography-assisted virtual endoscopy be an innovative tool for detecting urethral tissue pathologies? *Urology* 2014;83(4):930–8.

70. Behrens A, Heisterklaus I, Muller Y, et al. 2-D and 3-D visualization methods of endoscopic panoramic bladder images. Medical imaging 2011: visualization, image-guided procedures, and modeling. In: Proceedings of SPIE, vol. 7964. 2011. <http://proceedings.spiedigitallibrary.org/volume.aspx?volumeid=163#1599SessionName>.
71. Heinz-Peer G, Happel B, Memarsadeghi M, et al. Virtual multislice computed tomography cystoscopy for evaluation of urinary bladder lesions. *Radiologe* 2005; 45(10):897–904.
72. Russell ST, Kawashima A, Vrtiska TJ, et al. Three-dimensional CT virtual endoscopy in the detection of simulated tumors in a novel phantom bladder and ureter model. *J Endourol* 2005;19(2):188–92.
73. Stenzl A, Frank R, Eder R, et al. 3-dimensional computerized tomography and virtual reality endoscopy of the reconstructed lower urinary tract. *J Urol* 1998; 159(3):741–6.
74. Suleyman E, Yekeler E, Dursun M, et al. Bladder tumors: virtual MR cystoscopy. *Abdom Imaging* 2006;31(4):483–9.
75. Tidwell AS. Principles of computed tomography and magnetic resonance imaging. In: Thrall DE, editor. *Textbook of veterinary diagnostic radiology*. 5th edition. St Louis (MO): Saunders Elsevier; 2007. p. 50–77.
76. Samii VF. Inverted contrast medium-urine layering in the canine urinary bladder on computed tomography. *Vet Radiol Ultrasound* 2005;46(6):502–5.
77. Léveillé R. Ultrasonography of urinary bladder disorders. *Vet Clin North Am Small Anim Pract* 1998;28(4):799–821.
78. Scrivani PV, Chew DJ, Buffington CA, et al. Results of double-contrast cystography in cats with idiopathic cystitis: 45 cases (1993–1995). *J Am Vet Med Assoc* 1998;212(12):1907–9.
79. Zotti A, Fant P, De Zan G, et al. Chronic cystitis with ossification of the bladder wall in a 6-month-old German shepherd dog. *Can Vet J* 2007;48(9):935–8.
80. Aizenberg I, Aroch I. Emphysematous cystitis due to *Escherichia coli* associated with prolonged chemotherapy in a non-diabetic dog. *J Vet Med B Infect Dis Vet Public Health* 2003;50(8):396–8.
81. Middleton DJ, Lomas GR. Emphysematous cystitis due to *Clostridium perfringens* in a non-diabetic dog. *J Small Anim Pract* 1979;20(7):433–8.
82. Moon R, Biller DS, Smee NM. Emphysematous cystitis and pyelonephritis in a nondiabetic dog and a diabetic cat. *J Am Anim Hosp Assoc* 2014;50(2): 124–9.
83. Petite A, Busoni V, Heinen MP, et al. Radiographic and ultrasonographic findings of emphysematous cystitis in four nondiabetic female dogs. *Vet Radiol Ultrasound* 2006;47(1):90–3.
84. Martinez I, Mattoon JS, Eaton KA, et al. Polypoid cystitis in 17 dogs (1978–2001). *J Vet Intern Med* 2003;17(4):499–509.
85. Takiguchi M, Inaba M. Diagnostic ultrasound of polypoid cystitis in dogs. *J Vet Med Sci* 2005;67(1):57–61.
86. Polzin DJ, Jeraj K. Urethritis, cystitis, and ureteritis. *Vet Clin North Am Small Anim Pract* 1979;9(4):661–78.
87. Moroff SD, Brown BA, Matthiesen DT, et al. Infiltrative urethral disease in female dogs: 41 cases (1980–1987). *J Am Vet Med Assoc* 1991;199(2):247–51.
88. Langston C, Gisselman K, Palma D, et al. Diagnosis of urolithiasis. *Compend Contin Educ Vet* 2008;30(8):447–50, 452–4; [quiz: 455].
89. Voros K, Wladar S, Vrabely T, et al. Ultrasonographic diagnosis of urinary bladder calculi in dogs. *Canine Pract* 1993;18(1):29–33.

90. Weichselbaum RC, Feeney DA, Jessen CR, et al. Relevance of sonographic artifacts observed during in vitro characterization of urocystolith mineral composition. *Vet Radiol Ultrasound* 2000;41(5):438–46.
91. Byl KM, Kruger JM, Kinns J, et al. In vitro comparison of plain radiography, double-contrast cystography, ultrasonography, and computed tomography for estimation of cystolith size. *Am J Vet Res* 2010;71(3):374–80.
92. Knapp DW. Tumors of the urinary system. In: Withrow SJ, Vail DM, editors. *Small animal clinical oncology*. 4th edition. St Louis (MO): Saunders Elsevier; 2007. p. 649–58.
93. Confer AW, Panciera RJ. The urinary system. In: McGavin MD, Carlton WW, Zachary JF, editors. *Thomson's special veterinary pathology*. 3rd edition. St Louis (MO): Mosby; 2001. p. 235–77.
94. Léveillé R, Biller DS, Partington BP, et al. Sonographic investigation of transitional cell carcinoma of the urinary bladder in small animals. *Vet Radiol Ultrasound* 1992;33(2):103–7.
95. Hanazono K, Fukumoto S, Endo Y, et al. Ultrasonographic findings related to prognosis in canine transitional cell carcinoma. *Vet Radiol Ultrasound* 2014; 55(1):79–84.
96. Heng HG, Lowry JE, Boston S, et al. Smooth muscle neoplasia of the urinary bladder wall in three dogs. *Vet Radiol Ultrasound* 2006;47(1):83–6.
97. Benigni L, Lamb CR, Corzo-Menendez N, et al. Lymphoma affecting the urinary bladder in three dogs and a cat. *Vet Radiol Ultrasound* 2006;47(6):592–6.
98. Ticer JW, Spencer CP, Ackerman N. Transitional cell carcinoma of the urethra in 4 female dogs: its urethrographic appearance. *Vet Radiol* 1980;21(1):12–7.
99. Liptak J, Dernell WS, Withrow SJ. Haemangiosarcoma of the urinary bladder in a dog. *Aust Vet J* 2004;82(4):215–7.
100. Mellanby RJ, Chantrey JC, Baines EA, et al. Urethral haemangiosarcoma in a boxer. *J Small Anim Pract* 2004;45(3):154–6.
101. Olausson A, Stieger SM, Loefgren S, et al. A urinary bladder fibrosarcoma in a young dog. *Vet Radiol Ultrasound* 2005;46(2):135–8.
102. Mirkovic TK, Shmon CL, Allen AL. Urinary obstruction secondary to an ossifying fibroma of the os penis in a dog. *J Am Anim Hosp Assoc* 2004;40(2):152–6.
103. Suzuki K, Nakatani K, Shibuya H, et al. Vaginal rhabdomyosarcoma in a dog. *Vet Pathol* 2006;43(2):186–8.
104. Winter MD, Locke JE, Penninck DG. Imaging diagnosis—urinary obstruction secondary to prostatic lymphoma in a young dog. *Vet Radiol Ultrasound* 2006;47(6): 597–601.
105. Nyland TG, Wallack ST, Wisner ER. Needle-tract implantation following US-guided fine-needle aspiration biopsy of transitional cell carcinoma of the bladder, urethra, and prostate. *Vet Radiol Ultrasound* 2002;43(1):50–3.
106. Côté E, Carroll MC, Beck KA, et al. Diagnosis of urinary bladder rupture using ultrasound contrast cystography: in vitro model and two case-history reports. *Vet Radiol Ultrasound* 2002;43(3):281–6.
107. Stafford JR, Bartges JW. A clinical review of pathophysiology, diagnosis, and treatment of uroabdomen in the dog and cat. *J Vet Emerg Crit Care* 2013; 23(2):216–29.
108. Groesslinger K, Tham T, Egerbacher M, et al. Prevalence and radiologic and histologic appearance of vesicourachal diverticula in dogs without clinical signs of urinary tract disease. *J Am Vet Med Assoc* 2005;226(3):383–6.
109. Rahal SC. What is your diagnosis? *J Am Vet Med Assoc* 2004;225(7): 1041–2.

110. Agut A, Lucas X, Castro A, et al. A urethrorectal fistula due to prostatic abscess associated with urolithiasis in a dog. *Reprod Domest Anim* 2006;41(3):247–50.
111. Hage MC, Duarte TS, Tavares TR, et al. Radiographic diagnosis of traumatic urethrorectal fistula in dog. *Ciencia Rural* 2011;41(5):848–51.
112. Miller CF. Urethrorectal fistula with concurrent urolithiasis in a dog. *Vet Med Small Anim Clinician* 1980;75(1):73–6.
113. Kochakarn W, Pummanagura W. Foreign bodies in the female urinary bladder: 20-year experience in Ramathibodi Hospital. *Asian J Surg* 2008;31(3):130–3.
114. Henninger W. Air-gun pellet forming the nucleus of a cystic calculus in a dog. *Wien Tierarztl Monatsschr* 1998;85(11):395–8.
115. Houston DM, Eaglesome H. Unusual case of foreign body-induced struvite urolithiasis in a dog. *Can Vet J* 1999;40(2):125–6.
116. Wyatt KM, Marchevsky AM, Kelly A. An enterovesical foreign body in a dog. *Aust Vet J* 1999;77(1):27–9.